CONCEPTS IN AGENTIC AI

AI for Design, ISDN 3150, 2026 Spring

RECAP

- Agent definition

  -Agent

  -Agentic Workflow

  -Multi-Agent System

- Agentic Workflow Design

  -General design steps

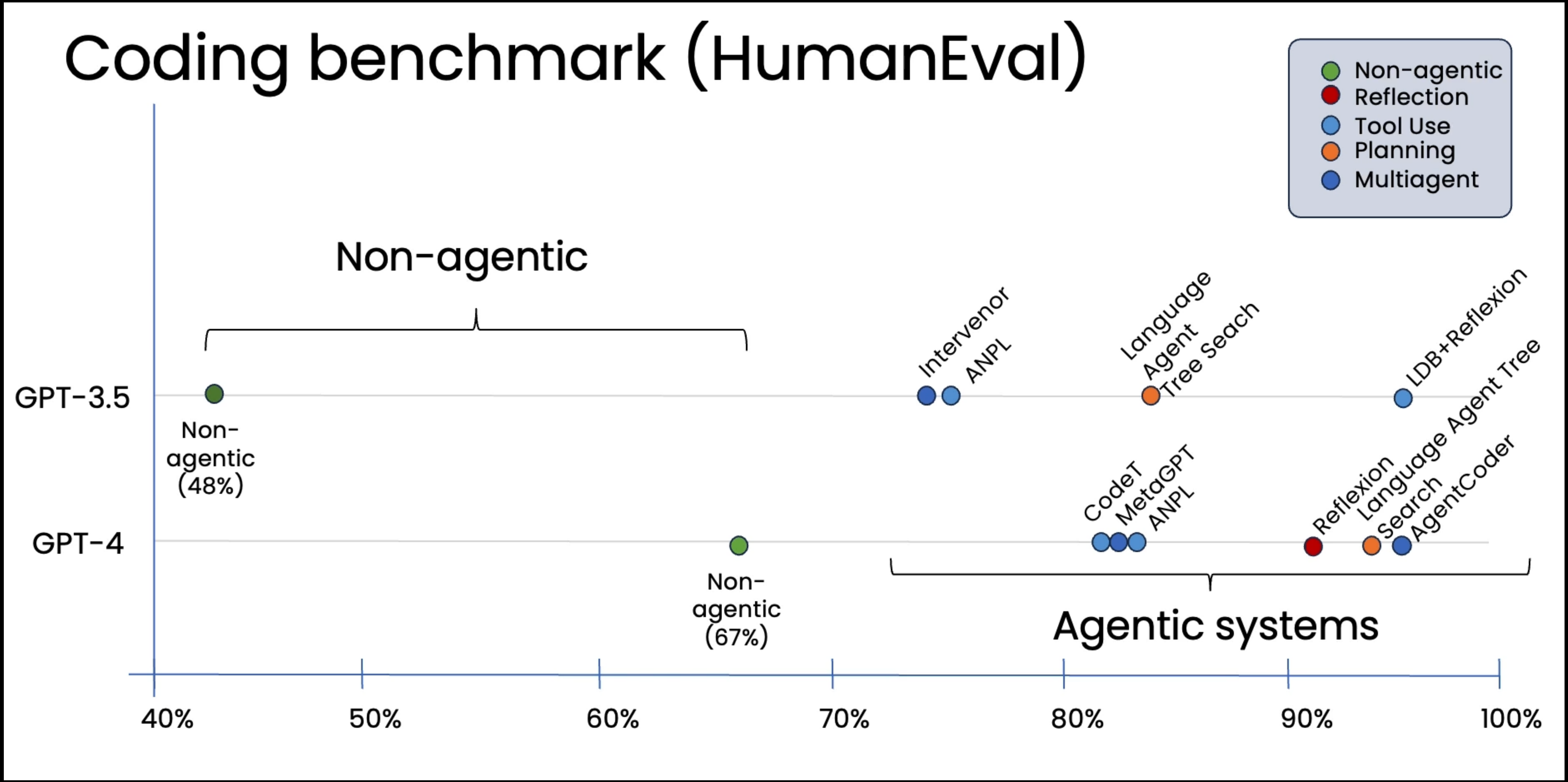
  -Reflection

  -Tool using

  -Planning

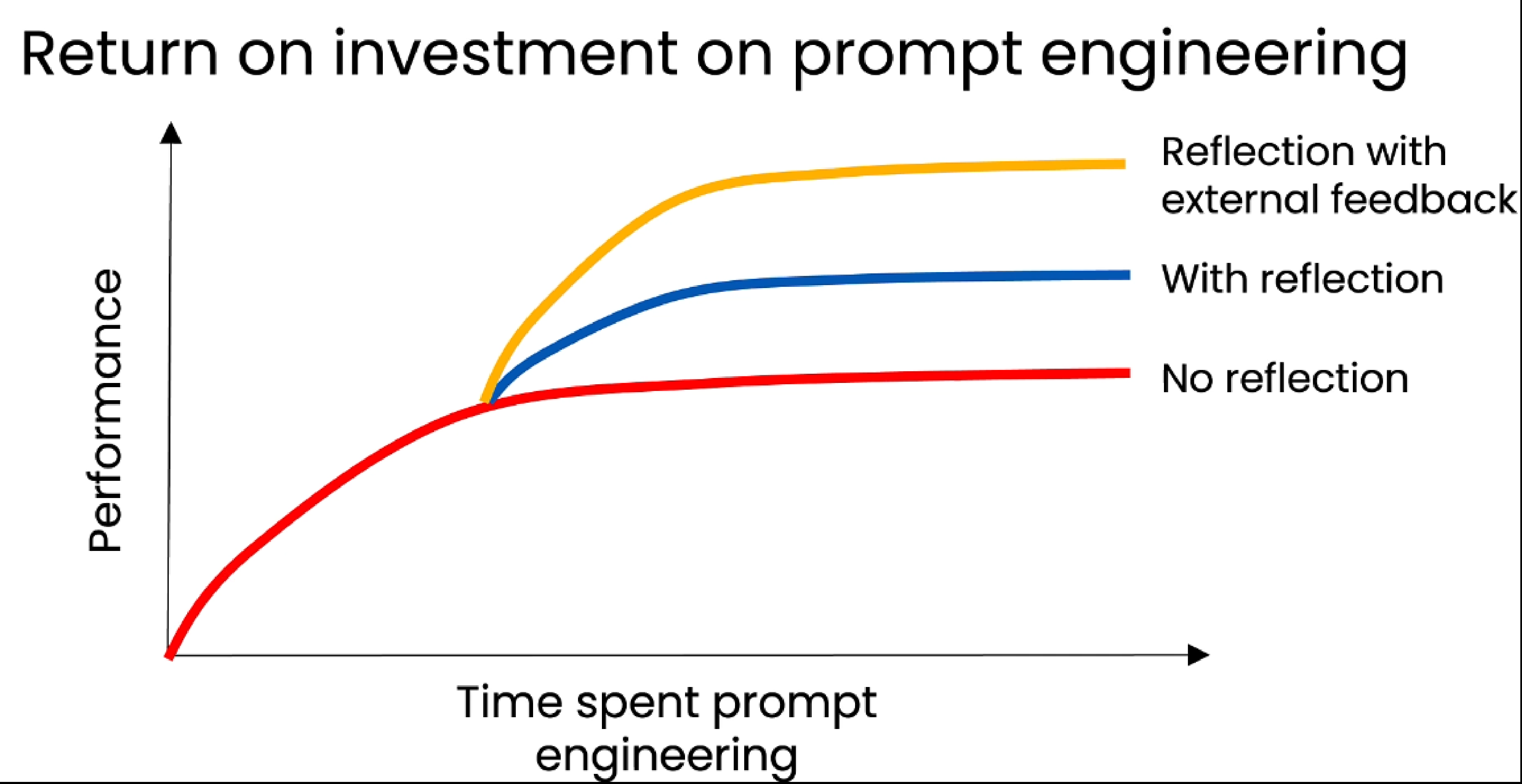
  -Multi-agent planning

WHY AGENTIC AI?

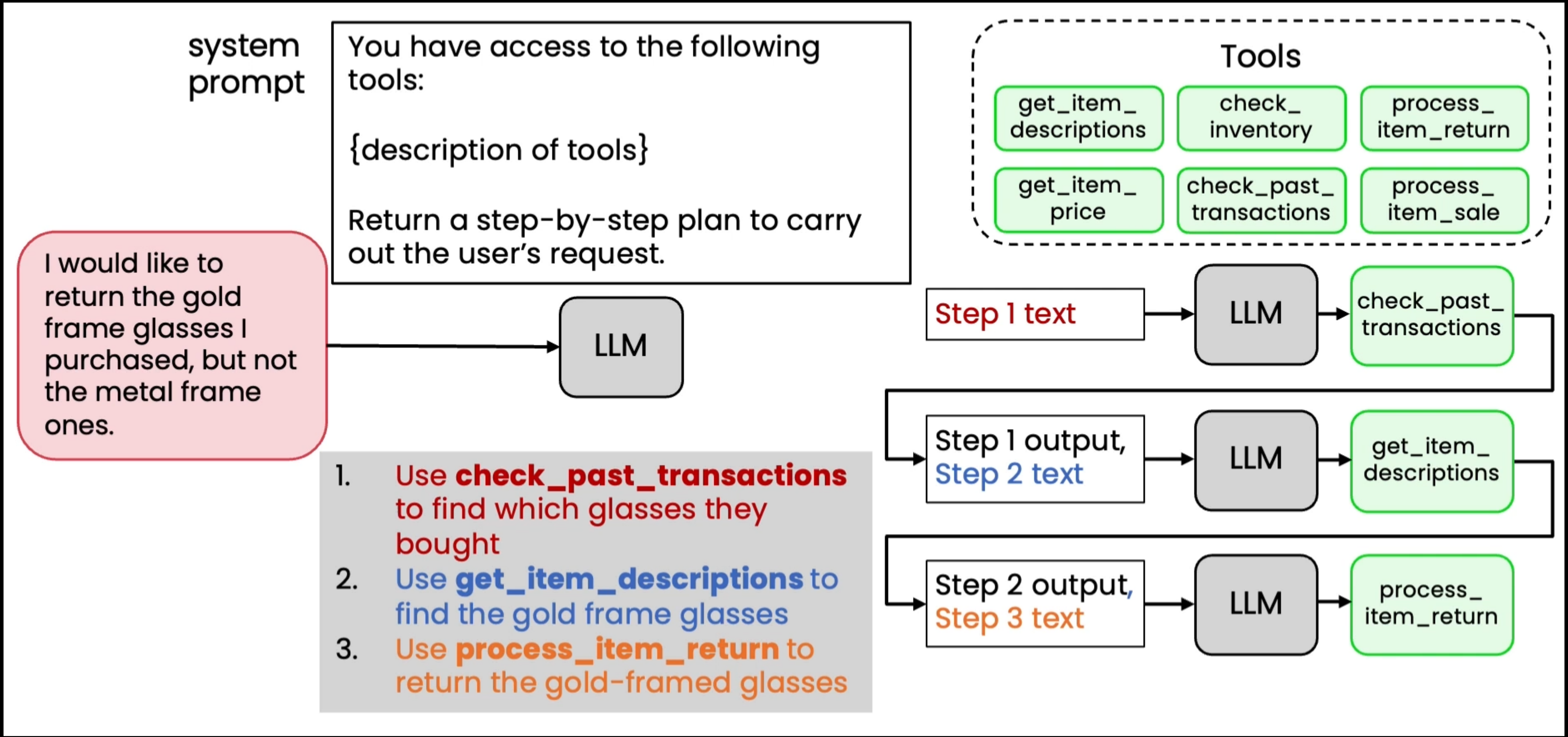


REFLECTION IS HELPFUL!

- Normal prompting → self-reflection → external reflection

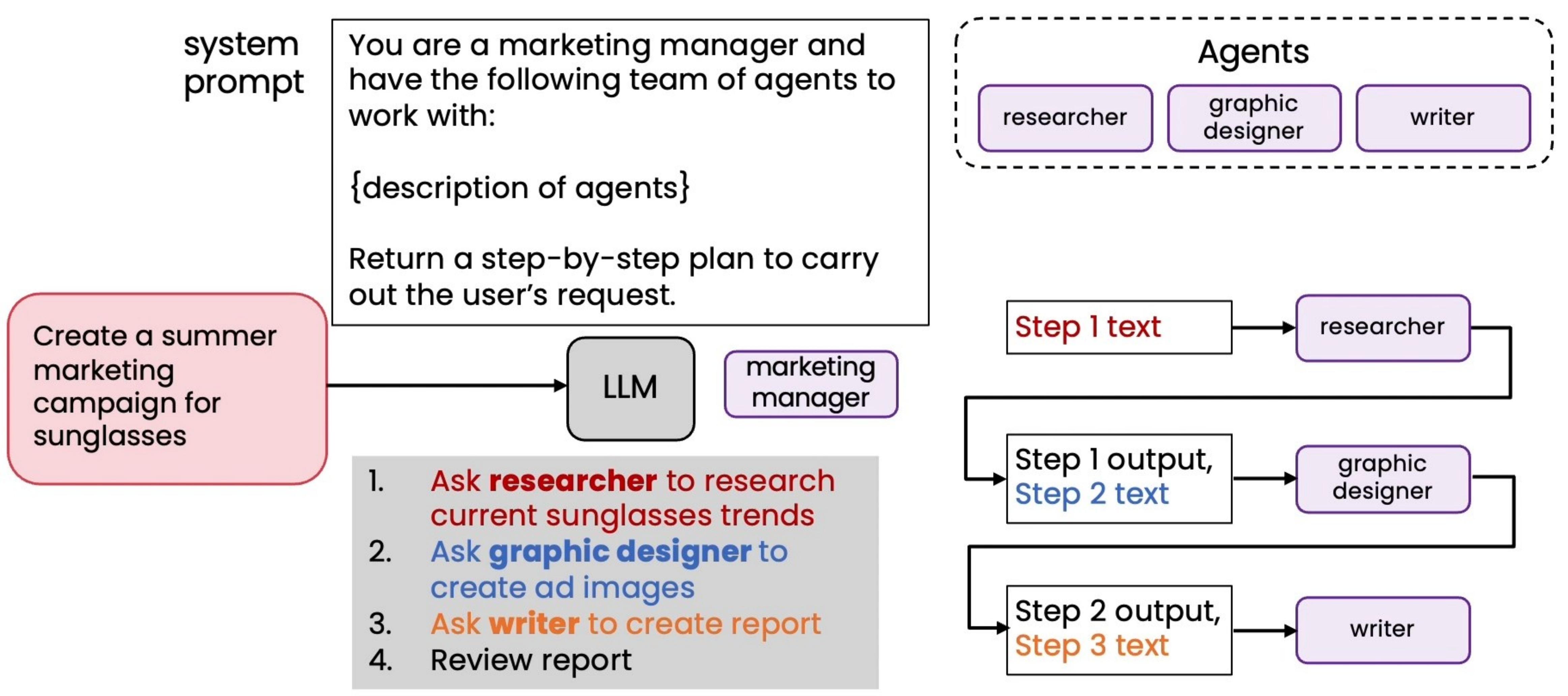


PLANNING DESIGN PATTERN



MULTI-AGENTIC WORKFLOWS PLANNING

- Planning with agents

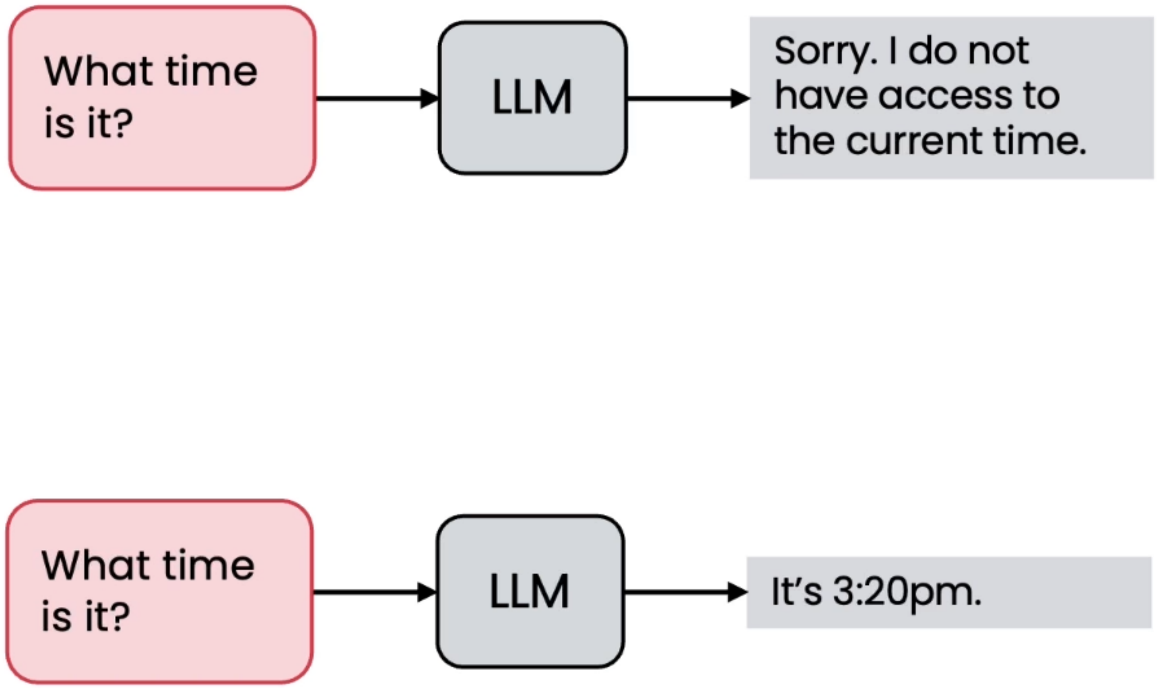


INDEX

- What is MCP?
- What is Skills?

MODEL CONTEXT PROTOCOL (MCP)

- Background:
- Function calling in LLMs

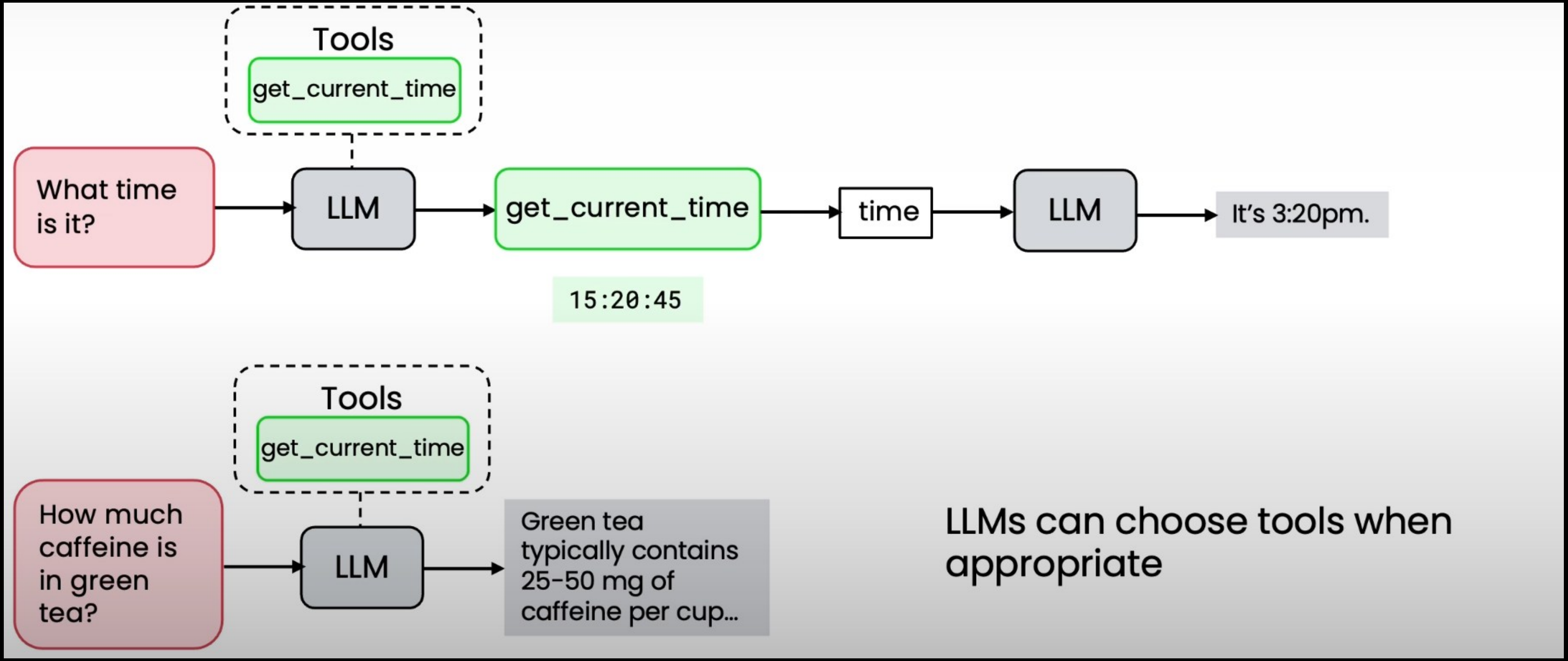


get_current_time() function:

```python
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""


    return datetime.now().strftime("%H:%M:%S")
```

MODEL CONTEXT PROTOCOL (MCP)

- Background:
- Function calling in LLMs

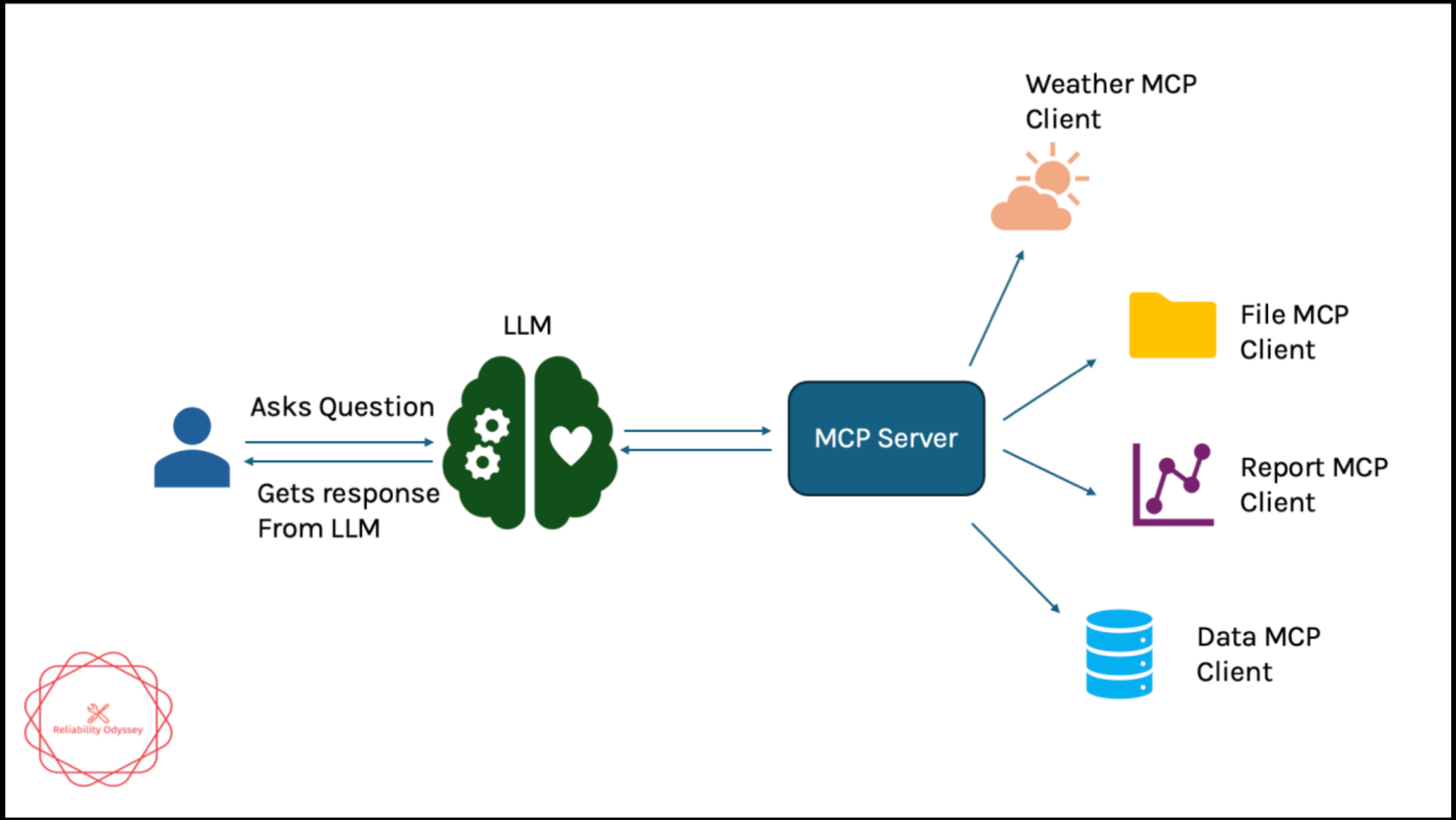


MODEL CONTEXT PROTOCOL (MCP)

- MCP: a set of self-explained APIs for LLMs to use.
- **MCP** is an open-source protocol designed to replace the fragmented, "one-off" integrations between AI models and data sources. Before MCP, connecting an AI to a specific database required writing custom glue code for every different model or framework.
- With MCP, you write a **Server** once, and it becomes immediately accessible to any **Host** (like Claude Desktop, an IDE, or a custom-built autonomous agent).

MODEL CONTEXT PROTOCOL (MCP)

- **The Client:** The component within the Host that maintains the connection to the server.
- **The Server:** A lightweight program that exposes specific capabilities. It doesn't just provide "functions"; it provides a structured context.



HOW IT WORKS: THE "INTROSPECTION" FLOW

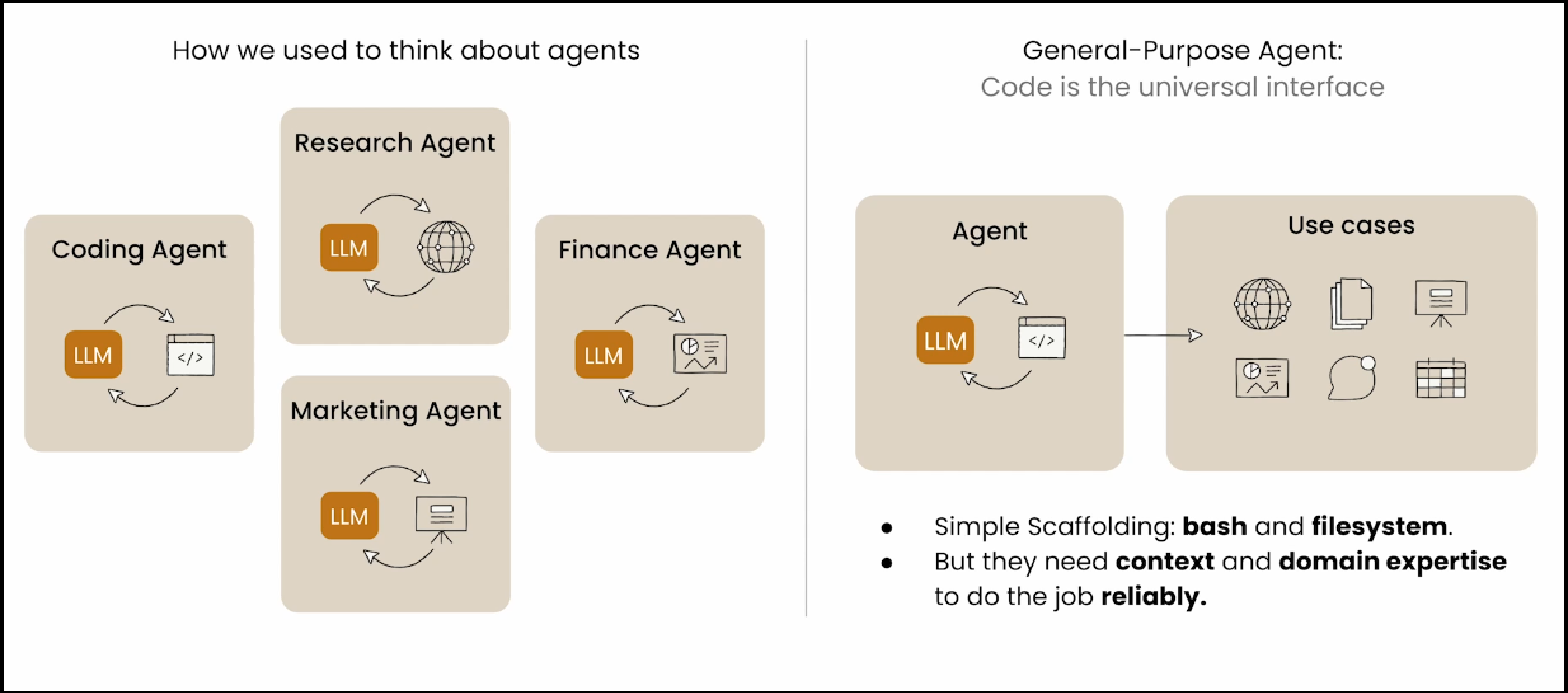
- **Handshake:** The Host starts the MCP Server.
- **Self-Description:** The Server sends a "menu" to the Host. It says: *"I have a tool called query_3dgs_metadata and a resource called research_notes.md."*
- **Reasoning:** When a user asks a question, the model looks at this menu and decides which tool or resource fits the intent.
- **Execution:** The Host executes the call locally (often via stdio or HTTP/SSE), fetches the result, and feeds it back to the model's context.

WHAT IS AN AGENT SKILL?

- Questions: How can we share Agents?
- Share the prompt?

WHAT IS AN AGENT SKILL?

- General Purpose Agent

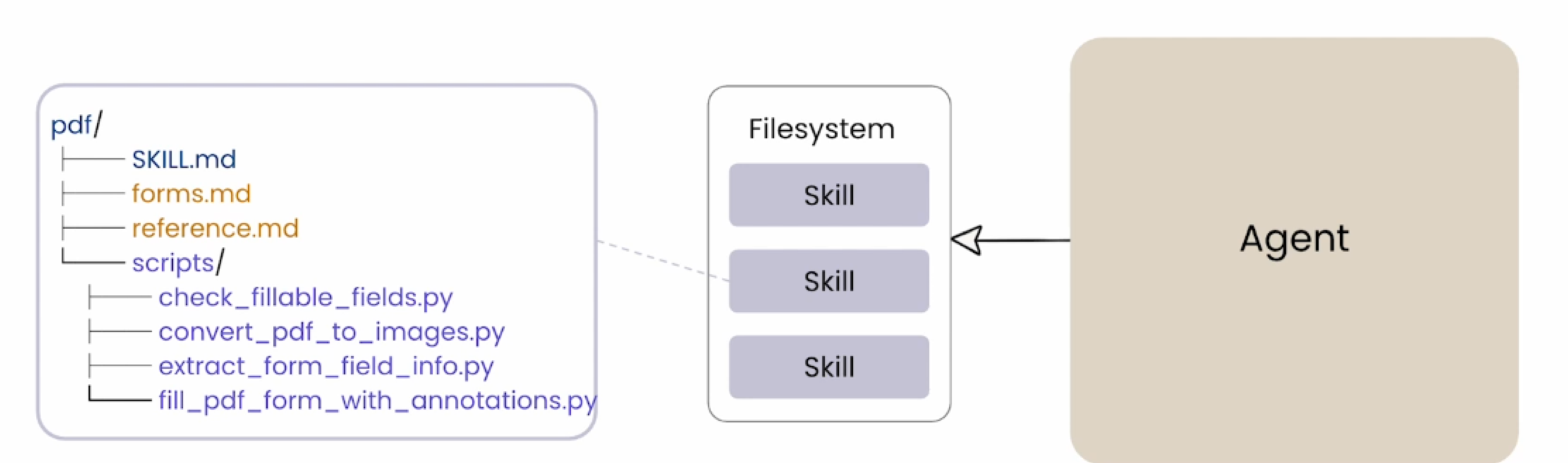


WHAT IS AN AGENT SKILL?

- A skill is more like a workflow containing

  Context information

  Functions (tools)



CONTEXT MANAGEMENT OF USING SKILLS

analyzing-marketing-campaign/SKILL.md

YAML Frontmatter

```
---
name: analyzing-marketing-campaign
description: Analyze weekly marketing campaign performance data
across channels. Use when analyzing multi-channel digital
marketing data to calculate funnel metrics and compare to
benchmarks, compute cost and revenue efficiency metrics, or get
budget reallocation recommendations based on performance rules.
---
```

Metadata: always loaded

Markdown

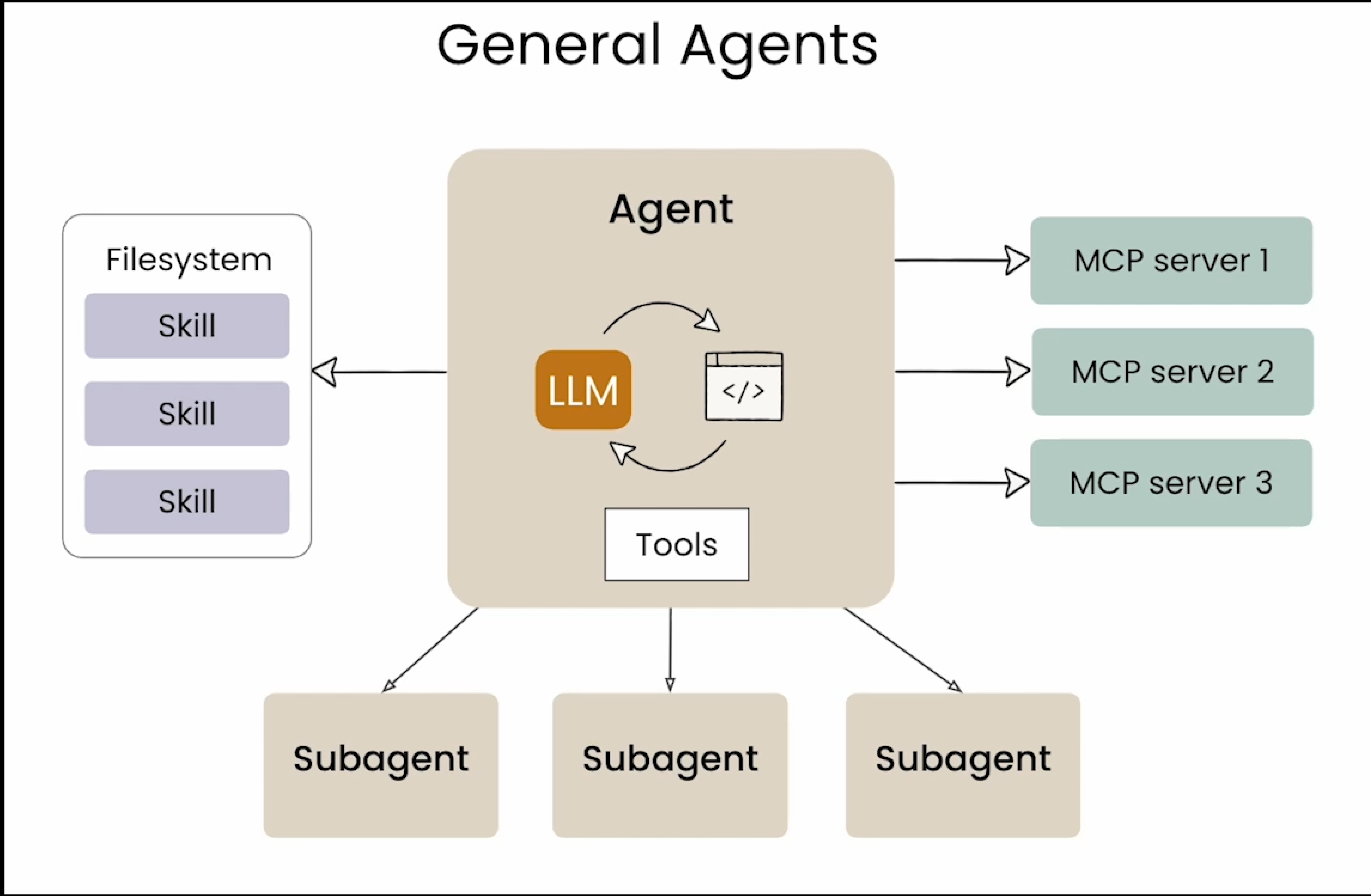
```
## Input

Excel/CSV with columns: **Date, Campaign_Name, Channel, Segment, Impressions,
Clicks, Conversions, Spend, Revenue, Orders**

## Funnel Metrics (per channel)

- **CTR** = Clicks / Impressions × 100
- **CVR** = Conversions / Clicks × 100

## Efficiency Metrics (per channel)

- **ROAS** = Revenue / Spend (target: ≥4.0x)
- **CPA** = Spend / Conversions (max: $50)
- **Net Profit** = Revenue - (Spend + Orders×$8 + Revenue×35%)

## Output Tables

**Funnel:** | Channel | CTR Actual | CTR Benchmark | CTR Diff | CVR Actual | CVR
Benchmark | CVR Diff |
**Efficiency:** Channel | ROAS | Status | CPA | Status | Net Profit | Status |
Include brief interpretation and recommendations per channel

## Budget Reallocation

If requested, see `references/budget_reallocation_rules.md` for decision
framework.
```

Instructions: loaded when triggered

Resources: loaded as needed

```
## Weekly Optimization Framework

### Rule 1: Performance-Based Decision Framework
Use the following criteria to determine budget adjustments for each marketing
channel on a weekly basis.

**Increase Budget by 10-15%**

A channel qualifies for budget increase when it meets all of the following
conditions:
- ROAS exceeds the target threshold by 15% or more

... and so on ...
```

SKILLS, TOOLS, MCP



FINAL PROJECT PRESENTATION DATE

- Remember to vote for the date of final project presentation!
- https://docs.google.com/forms/d/e/1FAIpQLSd9KgZhD5aiq34Nq4qiFc1sIMqI3VNEXs1q6M2kpY3BVoLf4Q/viewform?usp=dialog

THANK YOU!

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